

EPGSpeller: Open-Vocabulary Silent Spelling with Electropalatography

Anonymous Authors

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Abstract

We study open-vocabulary (lexicon-free, character-level) silent spelling from electropalatography (EPG) under a word-holdout protocol. We ask whether injecting a fixed proxy electrode layout as a spatial inductive bias improves character-level decoding. We compare (i) a layout-agnostic channel vector front-end, (ii) a proxy-grid projection consumed by a two-dimensional convolutional front-end, and (iii) row/column pooling on the same proxy grid. In this setting, the proxy-grid convolutional front-end has higher (worse) greedy CER than the vector baseline and a higher (worse) RTF, while proxy-grid spatial augmentation reduces CER degradation under fixed electrode dropouts.

1 Problem setting

Electropalatography (EPG) measures tongue–palate contact as a multichannel time series from a fixed electrode array. EPG has long been used to study and visualize tongue–palate contact patterns in phonetic research and speech training (e.g., [9, 10, 16]), and prior work has explored EPG data reduction and representation [11, 2]. We work with a fixed inventory of 124 electrode channels indexed by the SmartPalate channel numbering (SmartPalate refers to the EPG device and its channel indexing scheme), and we use that index order as the canonical channel order for layout-agnostic models.

To introduce spatial structure, we define a fixed proxy mapping π from SmartPalate channel indices to grid coordinates on a 16×16 grid, derived from the pinned SmartPalate distribution snapshot

`ev_scripts_smartpalate_distribution_csv` (`d21d93c9ec3c`). Operationally, the snapshot is a 16×16 table of channel indices; $\pi(c)$ is the first row/column location of channel index c in that table. Channels absent from the snapshot are treated as *unmapped* and retained as separate non-grid features (4); 5 channel indices appear multiple times in the snapshot and are resolved deterministically by first occurrence. We also provide a separate binary mask channel indicating electrode presence on the proxy grid. This proxy mapping is used only as an inductive bias for layout-aware representations; we do not interpret it as a geometric tongue model.

We use a character-level connectionist temporal classification (CTC) formulation [8]. In the word-holdout protocol, training and test sets are disjoint at the word-identity level, where word identity is defined as the normalized target-word string used for evaluation; all examples of each test word are excluded from training. We decode with greedy CTC and do not apply a lexicon constraint or an external language model.

2 Related work and positioning

Silent speech interfaces (SSI) decode linguistic content from non-acoustic signals; surveys summarize modality taxonomies and highlight protocol dependence across systems [3, 5, 7, 14]. SSI has been explored with multiple sensing modalities, including ultrasound and optical tongue imaging [12] and implanted magnetic-sensor setups [6]; here we focus on electropalatography (EPG), whose palatogram-style structure motivates explicit spatial representations and reduction analyses [16, 11, 2, 17].

Closest to our task, SilentSpeller studies EPG-based silent spelling and text entry [13], while ReHEarSSE provides a recent non-EPG silent-spelling reference point [4]; EPG has also been used in multimodal learning beyond text entry [1]. In this work, under a fixed word-holdout protocol and greedy CTC decoding, we isolate the effect of proxy-layout-aware front-ends (proxy-grid two-dimensional convolution or row/column compression) and evaluate robustness to missing electrodes via structured masking and augmentation [15]. We therefore scope claims to this protocol and proxy-layout assumption, and do not claim new hardware, a new electrode geometry, or a new decoding objective.

3 Models and spatial front-ends

All variants share the same temporal backbone and CTC decoding setup. The temporal backbone produces 64-dimensional per-frame embeddings and uses the same recurrent stack across conditions (uni-gru; 5layers; 512hidden units). We apply a temporal unfolding operator (sliding-window chunking) with kernel length 32 and stride 4 before the recurrent stack; we vary only the EPG spatial front-end described below. For budgeted settings, we restrict the input to a subset of size K ; for each K , the same selected channels are provided to all compared spatial front-ends.

Vector (layout-agnostic). The vector baseline uses the canonical channel index order and does not use any electrode layout information.

Layout-aware two-dimensional reconstruction. We scatter the selected channels into the 16×16 proxy grid and apply a two-dimensional convolutional front-end to obtain per-frame embeddings. Grid cells without electrodes are masked, duplicated channels are aggregated deterministically, and unmapped channels are embedded separately and combined with the grid embedding.

Row/column compression. We form row-wise and column-wise pooled summaries from the same proxy grid, concatenate them with the unmapped channels, and project to the shared embedding size.

4 Experimental protocol

We evaluate the word-holdout protocol using 4 random split seeds (seed0, seed1, seed2, seed3) and reuse the exact same splits across all compared spatial front-ends. Training uses 10000 optimization steps at 10.0 ms frame shift and enables SpecAugment-inspired masking [15].

We compare two electrode-selection baselines for budgeted subsets of size K : *TopK*, which selects the K highest-ranked channels under an occlusion-based importance score (performance degradation induced by masking a channel), and *FPS-diverse TopK*, which adds a proxy-grid spatial-diversity constraint via farthest-point sampling from a high-scoring candidate pool. We treat both as baselines and do not claim novelty for these heuristics.

We report character error rate (CER) from greedy CTC decoding and streaming real-time factor (RTF). CER is computed as total edit distance divided by total reference characters on the evaluation set. RTF is computed as total model forward-pass wall time divided by total input duration (derived from the number of input frames and the 10.0 ms frame shift). Lower is better for both metrics.

5 Results

Scope. All comparisons below use the same protocol and matched training settings; we interpret the findings as evidence about proxy-layout spatial inductive biases under this setting, not as claims about true geometric tongue–palate structure.

5.1 Two-dimensional convolution vs vector

Table 1 compares vector and two-dimensional convolutional front-ends for multiple electrode budgets. In this protocol and setting, the proxy-grid convolutional front-end has higher (worse) greedy CER than the vector baseline and a higher (worse) RTF.

5.2 Row/column compression vs vector

Table 2 shows that row/column compression yields similar greedy CER to the vector baseline, but a higher (worse) RTF.

Table 1: Vector vs two-dimensional convolution (mean $\pm$ std over splits). Evidence: ev_ed20260217_h11_spatial2d_vs_vector_plus3arekACER96relative4to2026-evaluation. (2f2840f6dee5).

K	Vector Greedy CER	Spatial2D Greedy CER	Vector Stream RTF	Spatial2D Stream RTF
TopK				
32	0.1399 $\pm$ 0.0162	0.4885 $\pm$ 0.1269	0.000491 $\pm$ 0.000035	0.002670 $\pm$ 0.000127
64	0.1289 $\pm$ 0.0135	0.2609 $\pm$ 0.0300	0.000476 $\pm$ 0.000004	0.002895 $\pm$ 0.000304
96	0.1194 $\pm$ 0.0173	0.2304 $\pm$ 0.0350	0.000475 $\pm$ 0.000023	0.002767 $\pm$ 0.000238
124	0.1048 $\pm$ 0.0093	0.1831 $\pm$ 0.0103	0.000469 $\pm$ 0.000014	0.002639 $\pm$ 0.000067
FPS2K				
32	0.1440 $\pm$ 0.0136	0.5795 $\pm$ 0.3052	0.000514 $\pm$ 0.000022	0.002674 $\pm$ 0.000129
64	0.1195 $\pm$ 0.0103	0.3581 $\pm$ 0.3079	0.000532 $\pm$ 0.000094	0.002808 $\pm$ 0.000180
96	0.1137 $\pm$ 0.0161	0.3438 $\pm$ 0.3104	0.000486 $\pm$ 0.000029	0.002626 $\pm$ 0.000096
124	0.1048 $\pm$ 0.0093	0.1831 $\pm$ 0.0103	0.000469 $\pm$ 0.000014	0.002639 $\pm$ 0.000067

Table 2: Vector vs row/column compression (mean $\pm$ std over splits). Evidence: ev_ed20260217_h13_rowcol_vs_vector_pl_k32_k64_2026-02-17. (a7fb56ec370d).

K	Vector Greedy CER	RowCol Greedy CER	Vector Stream RTF	RowCol Stream RTF
TopK				
32	0.1399 $\pm$ 0.0162	0.1448 $\pm$ 0.0204	0.000491 $\pm$ 0.000035	0.002141 $\pm$ 0.000276
64	0.1289 $\pm$ 0.0135	0.1269 $\pm$ 0.0143	0.000476 $\pm$ 0.000004	0.002072 $\pm$ 0.000142
FPS2K				
32	0.1440 $\pm$ 0.0136	0.1466 $\pm$ 0.0080	0.000514 $\pm$ 0.000022	0.002176 $\pm$ 0.000311
64	0.1195 $\pm$ 0.0103	0.1255 $\pm$ 0.0193	0.000532 $\pm$ 0.000094	0.002104 $\pm$ 0.000216

5.3 Spatial augmentation and robustness to electrode failures

To quantify robustness, we apply fixed electrode-dropout masks at evaluation time (the same masked channels for all test samples in an evaluation run) and report the increase in greedy CER relative to clean evaluation. Here, *spatial augmentation* refers to training-time masking or perturbations that operate on the proxy electrode grid (rather than treating channels as exchangeable). Table 3 shows that spatial augmentation reduces the CER degradation under dropout.

6 Artifacts and auditability

This project maintains a claim-evidence registry in `paper/paper.json`. All numeric facts and result tables in this draft are generated from pinned evidence snapshots via `scripts/paper/generate_facts_tex.py` using `paper/facts_spec.yaml`. A strict checker rejects handwritten numeric facts in the manuscript

Table 3: Fixed-dropout robustness for the two-dimensional front-end (mean $\pm$ std over splits). Values are CER relative to clean evaluation. Evidence: ev_ed20260217_h12_spatial_aug_fixed_dropout_summary (0cd1fecedc16).

Drop rate	No spatial aug	Spatial aug
0.0	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
0.1	0.0245 $\pm$ 0.0027	0.0183 $\pm$ 0.0065
0.2	0.0783 $\pm$ 0.0162	0.0660 $\pm$ 0.0097
0.3	0.1440 $\pm$ 0.0183	0.1224 $\pm$ 0.0120

body. To regenerate the manuscript numbers from a clean checkout, run: `python3scripts/paper/generate_facts_tex.py--facts_specpaper/facts_spec.yaml--paper_jsonpaper/paper.json--root.` and `python3scripts/paper/check_no_manual_numeric_facts.py--texmain.tex--facts_specpaper/facts_spec.yaml--root..` To audit claim-evidence consistency, run: `python3tools/paper_json_lint.py--paper-jsonpaper/paper.json--root.--strict` and `python3scripts/render_paper_json_claim_audit.py--paper-jsonpaper/paper.json--root.--out-mdpaper/snippets/claim_audit.md--strict.`

Key pinned evidence identifiers used by this draft include `ev_scripts_smartpalate_distribution_csv (d21d93c9ec3c)`, `ev_ed20260217_h11_spatial2d_vs_vector (2f2840f6dee5)`, `ev_ed20260217_h13_rowcol_vs_vector_pl (a7fb56ec370d)`, and `ev_ed20260217_h12_spatial_aug_fixed_dropout_summary (0cd1fecedc16)`.

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